# AEGIE: A Research Framework for Adaptive Agentic AI-Based Voice Interview Evaluation

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*Abstract– We describe AEGIE, an adaptive framework that is designed to interpret candidate answers to technical interview questions. This framework takes as input the candidates' answer in speech, takes account of the content of the answer, and also certain features of delivery. Speech transcription is provided by NVIDIA Parakeet; other measures reported separately include speaking rate, filler words, pauses, and related speech features. Text responses are interpreted via the MiniLM comparison to reference knowledge, at the level of whole answer, individual sentence, and expected concepts. The framework is instantiated with three functional agents that report on evidence, adapt the question, and display evidence. A previous framework step to be "question difficulty adaption" is implemented using fixed threshold decision rather than a learned classification. A "strawman" three question development test indicated that the module could be combined into the full process, from capturing a voice sample to transcription, semantic matching, adaptive question selecting, and evidence displaying; and the measurements at this stage are better seen as prototype observations, and the no-cost indication of candidate ability, than as validated levels for adaptive procedure. The next phase will involve responses from users and ratings from independent expert raters to calibrate the framework.*

*Keywords— Adaptive interviewing, Agentic AI, Automatic speech recognition, MiniLM, Semantic similarity, Technical interview evaluation, Voice analysis.*

## I. INTRODUCTION

Technical interview will be used to verify a candidate's knowledge in the technical field and his conception of the solution for a problem. A human interview can evaluate multiple ways that the same concept is said but becomes more challenging when many interviews must be led.

Both technical content and the delivery of a spoken response can be inferred from speech. Speech recognition may be used to transcribe the candidate's spoken answer, and characteristics such as speed of speech, pauses, and filler words may be used to describe the delivery. Speech measures do not, however, provide information about whether the expected technical concepts exist in an answer. Thus the transcript must be examined at a semantic level.

The AEGIE system accommodates this need by means of a flexible, voice-based interview. In response to the recorded reply, NVIDIA Parakeet transforms that into text and the system then distinguishes measurable speech properties. MiniLM is employed to match the candidate's response against known reference material but instead of using a single similarity score, the system makes comparisons across the whole answer, sentence and concept levels. This allows for evidence to be examined on separate portions of the response.

The system employs three functional agents: The Evaluation Agent structures the speech and semantic evidence produced for each turn. The Adaptive Agent exploits the structured evidence to select the level of difficulty of the next question, and The Feedback Agent communicates the evidences to the user. For the current prototype, question adaptation is governed by fixed threshold-based rules instead of a trained decision model.

This paper focuses on the larger question of whether these components can form an end-to-end interview pipeline. The measures used in the development experiment are treated as prototype scores for now, until their validity as competency or hiring scores can be established. A future stage will incorporate

responses from real participants and independent expert annotations to tune the framework.

## II. RELATED WORK

Many studies have looked at various signals collected during the candidate interview. Rao et al. [1] examined communication skills, with lexical, acoustic and visual inputs, and Rasipuram and Jayagopi [2] utilized both verbal and non-verbal signals during interview-based interaction. Other research, by Nguyen et al. [3], considered the association between non-verbal behaviour and perceived hirability, and similar work using conversational video resumes [4]. Suen et al. [5] investigated automated communication skills and personality assessment based on human interviewer ratings as reference information.

However, more recent research has focused on methods that unify multiple stages of the interview process. [6] and [7] describe systems that combine speech processing, natural language processing, automated questioning, feedback and simulated interview scenarios. Multimodal techniques have also been explored for analysis of interview performance [8]. These efforts suggest that automated systems for providing interview support may be useful, although the metrics computed by these systems must be carefully considered.

Another aspect of semantic representation that is related to the framework being proposed is the comparison of sentence-level meaning, for which sentence-BERT [9] is an efficient method. MiniLM [10] and MiniLMv2 [11] minimize the computation needed for Transformer representations and render them useful as compact representations of languages when computers are unable to support high compute capabilities. Transformer models are also explored for automatic short-answer assessment [12]- [14] by matching free-form answers with expected answer content.

The effects on candidates have also been explored. Studies [15]-[18] have looked at how applicants react to asynchronous video interview processes and algorithmic assessment. More recent research has looked at the potential for generative AI to be used as part of the interview process [19], and there has also been research on fairness and bias in algorithmic hiring systems [20].

By integrating these research avenues, however, AEGIE adds the integration of local speech transcription, quantifiable delivery features, multi-tiered semantic evidence, and in-situ question selection in a single system. Instead of a comprehensive set of generated measurements which might be viewed as absolute decision of the fairest hiring outcome, the prototype retains them as observable measures which could be cross checked with another set of independent human measures in future validation experiments with real participants.

## III. PROPOSED AEGIE FRAMEWORK

The voice-based processing workflow at AEGIE requires that all responses to interview questions go through the following processing flow. First, the candidate records the response in the interview interface. Next, when the response is submitted, the speech is transcribed using the NVIDIA Parakeet transcription model. During the same process, AEGIE also gathers quantifiable delivery metrics about the response, such as length of response, speaking rate, fillers, and pauses.

MiniLM is then applied after transcription to test the semantic proximity between the candidate answer and the reference knowledge. We conduct the analysis at three different levels instead of a single similarity test. Whole answer-level compares the answer as a whole with the reference knowledge; sentence-level analysis compare parts of the answer with the knowledge; concept-level analysis judges the candidate sentence' proximity to the expected technical concepts. For each expected concept, we preserve the candidate sentence with the highest semantic relation as support information.

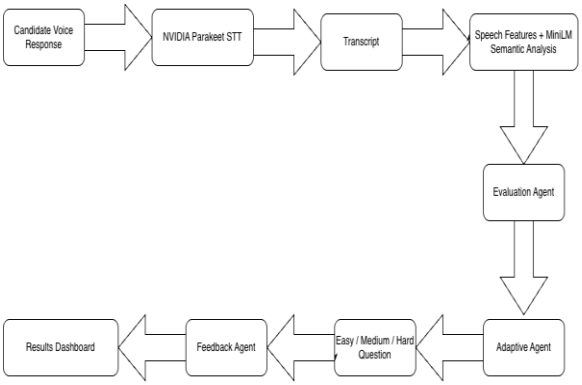



Fig. 1. Processing flow and agent structure used in the AEGIE prototype.

The information resulting from these calculations is handled by three agents; The Evaluation Agent gathers the speech measurements and semantic evidence generated for the current answer, The Adaptive Agent determines the next interview question's difficulty based on this data, and The Feedback Agent displays this processed evidence within the application interface providing the user access to the generated information.

The bank of interview questions is classified into Easy, Medium and Hard. The current implementation stage

imposes certain rules on movement between the levels. No adaptive decision model is trained at the moment. This rule based manner of adaptation makes it easier to track which questions are selected next while testing the prototype.

Another design consideration for AEGIE is that semantic evidence and delivery measures need to be preserved as two distinct types of information. For example, semantic similarity does not allow us to determine if a technical answer is completely correct, and delivery measures like pause frequency or speech rate do not directly characterize a technical skill set. That is why the model uses these metrics as evidence only for analysis and targeted follow-up questions, not as a basis for automated hiring decisions or pass/fail decisions.

## IV. RESEARCH METHODOLOGY AND DATA COLLECTION

The AEGIE research method has four sequential steps: collection of interview responses, speech treatment, extraction of semantic evidence and calibration by humans. The existing prototype performs all the automated phases. Acquisition of responses from more participants and quality assessment of the responses by independent experts will be conducted in the next validation phase.

### A. Interview Data Acquisition

In the interview scenario, the user speaks into a microphone and the technical question is shown on the application. The transcript received from the microphone can be linked to the participant id, question id, audio recording, transcript, difficulty level and evidence extracted. Keeping these elements together enables the measurements from framework to be traced back to the individual responses they were derived from.

### B. Speech Processing and Feature Extraction

The capture audio is passed through a model called NVIDIA Parakeet that transforms the audio of the response into text. At the same time, the framework logs perceptible metrics such as response length, number of words, speed of speech, use of filler words, and number of pauses. The speaking rate can be expressed in words per minute (WPM) as:

$$\text{WPM} = (\text{Number of spoken words} / \text{Response duration in seconds}) \quad (1)$$

These measurements are characteristics of candidate speech delivery. They are held as anecdotal data and are not expressed as a set of measures of technical knowledge and interview performance in and of themselves.

### C. Semantic Evidence Extraction

After that, we will tokenize the transcribed text and put the candidate response and reference content into a semantic space using MiniLM. Then we calculate the similarity value of the two content vectors. The method use cosine similarity to measure the relationship between the two:

$$\text{Similarity}(A,R) = (A.R) / (||A||\ ||R||), \quad (2)$$

Where A is the embedding of candidate's answer and R is the embedding of reference.

Instead of collapsing all measures of semantic similarity into one overall measure, AEGIE preserves evidence at the answer, sentence, and concept levels. At the concept level, we compare the target technical concept to the sentences that made up the candidate's response. The sentence that has the strongest semantic similarity to the concept is kept as evidence for the concept.

In a subsequent phase of the project, an independent human analysis will be added on the responses obtained from actual participants. The human annotators will review these responses to evaluate technical accuracy, pertinence, amount of content covered, and clarity of communication. Their annotations can be used to evaluate the evidence generated by AEGIE.

The data that is used for calibration will be stored apart from the data used for subsequent assessment. This is because it is desirable to have the calibrated framework tested against interview responses which were not used to perform the calibration, to give a more meaningful evaluation of its behaviour.

## V. ENVIRONMENTAL SETUP

The AEGIE prototype was developed as a web-based application running within a python flask framework that captured the broad spectrum of the interview process into a singular system. Voice recording, NVIDIA Parakeet for transcription, speech-related feature extraction, MiniLM semantic analysis, adaptive question selection and evidence presentation were developed as a single integrative application. An initial development interview consisting of five questions was conducted to establish if these modules could run as an end-to-end pipeline.

A. Development Interview Configuration

The set of interview questions is divided into three levels of difficulty - Easy, Medium, and Hard. Each question is linked to reference content and a list of the anticipated technical concepts. After a candidate submits a response, the framework evaluates the response and sends the evidence to the Adaptive Agent to use the current evidence to decide the next question's level of difficulty.

B. Prototype Evidence and Adaptation

In development experiment, for the concept level evidence, AEGIE combining the concept level evidence with the whole-answer semantic similarity to get a tentative evidence value. The calculation used in the prototype is:

$$E = 0.60C + 0.40W \quad (3)$$

Where C is the mean of concept-level similarities, and W is the whole-answer MiniLM similarity.

The evidence value produced is then classified into three provisional bands: scores below 0.45 are assigned the label 'Beginner', scores from 0.45 up to but not including 0.65 are assigned the label 'Intermediate', and scores of 0.65 and above are assigned the label 'Advanced'. These bands were developed solely for the purpose of testing the prototype, and have not been fine-tuned with respect to independently assigned participant responses. As such, the resulting labels should not be taken as confirmed candidate ability indicators.

C. Recorded Measures

The prototype saves certain measurements for each response as it is submitted. These measurements are: response time, number of words, word per minutes, number of filler words, number of pauses, whole-answer similarity, sentence-level similarity, concept-level evidence, and total processing time.

By storing the measurements separately, it is possible to follow the response as it is reconstructed by the framework. This also supplies the information necessary for a subsequent phase in the research in which the automated evidence can be compared with assessments from human experts.

## VI. RESULTS AND EVALUATION

The five-question development interview was used to test how the whole AEGIE pipeline operates. For each answer, the prototype transcribed the speech, took measurements of delivery, produced semantic evidence using MiniLM, selected the level of the next question, and showed the processed evidence. Because this was a single development run, the measurements mentioned in this experiment are of prototype behaviour, not of performance on a validated dataset of participants.

A. Speech Analysis Results

From the 5 responses, the average speed of the prototype was found to be 117.33 words/min. There was 1 filler and 10 pauses present. These results show that these measures (related to delivery) can be taken with the transcript and saved for later analysis. These measures are purely descriptive at this point and are not being analysed as measures of technical skill.

B. Semantic Evidence Results

The mean whole-answer MiniLM similarity between the five responses was 0.6522. At the concept level, 20 expected concepts were studied, with a mean best-match similarity of 0.4994. The whole-answer, sentence-level, and concept-level scores showed no similar trend over the individual questions; they all varied. This shows that they each provide a different perspective on the responses and they should each be kept as separate evidence during prototype assessment.

TABLE I QUESTION-WISE SEMANTIC AND SPEECH EVIDENCE

| Question | Whole MiniLM Semantic Similarity | Sentence MiniLM Semantic Similarity | Concept Mean Semantic Similarity | Total Processing Time |
|---|---|---|---|---|
| Q1. DS_E03 | 0.8460 | 0.6899 | 0.6937 | 8.211 s |
| Q2. DS_M02 | 0.8741 | 0.6437 | 0.5189 | 9.706 s |
| Q3. DS_H01 | 0.4366 | 0.4760 | 0.6089 | 8.339 s |
| Q4. DS_H02 | 0.6169 | 0.5300 | 0.4516 | 7.511 s |
| Q5. DS_H04 | 0.4874 | 0.3942 | 0.2237 | 7.959 s |

C. Provisional Semantic Evidence Level

Using the prototype evidence formula provided in Equation (3), the aggregate for the five responses would be 0.5605. According to the thresholds of current development, the subsequent level of the

category falls into the Intermediate. It is important to note that this scoring is solely used to test whether or not the adaptive feature functioned properly and should never be construed as a valid competency rating, interview score, recommendation to hire, or pass/fail.

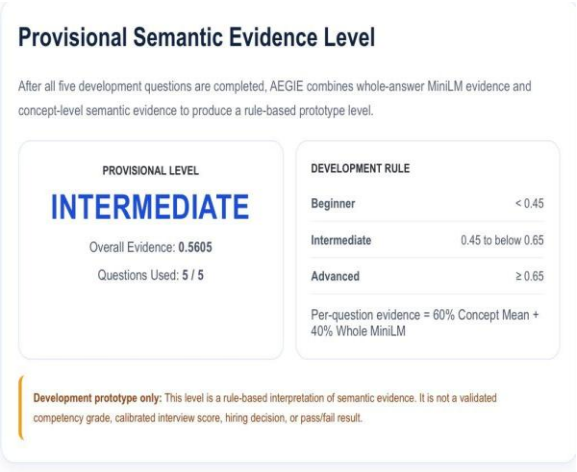



Fig. 2. Provisional semantic evidence level from the development prototype of AEGIE.

D. Processing Latency

Response time was also measured in order to track the behaviour of the prototype on the computational side. The speech-to-text calculations across the five responses took on average 4.717s with NVIDIA Parakeet. The Evaluation Agent and MiniLM semantic-analysis stage took on average 0.096s, and the complete server-side processing time took on average 8.345s per response. These times are representative of the current development setup and would vary when running the framework on different hardware/software.

TABLE II QUESTION-WISE PROCESSING LATENCY OF THE AEGIE PROTOTYPE

| Question | Parakeet STT Processing Time | Agent / MiniLM Processing Time |
|---|---|---|
| Q1. DS_E03 | 4.579 sec | 0.079 sec |
| Q2. DS_M02 | 5.709 sec | 0.124 sec |
| Q3. DS_H01 | 4.737 sec | 0.085 sec |
| Q4. DS_H02 | 4.138 sec | 0.109 sec |
| Q5. DS_H04 | 4.428 sec | 0.084 sec |

## VII. DISCUSSION AND LIMITATIONS

Development Testing showed that the core elements of AEGIE can function together in a single voice-based interview process. The prototype was able to take spoken responses and generate transcription, identified observable measures of delivery, and produce semantic evidence at the whole-answer, sentence, and concept levels. The semantic measurements varied by question across the five questions, suggesting that a single similarity score isn't enough to fully describe the response. Therefore, the current development version of AEGIE uses all three levels as evidence.

Concept-level Analysis adds more granularity by linking each predicted technical concept to the candidate sentence that has the highest semantic similarity to it. This is useful for reviewing what in the candidate response generated the evidence. Note, semantic similarity is not equated with correct or complete technical answers. Candidate responses can have semantically similar language without being a complete or correct technical response. As such, AEGIE considers these measurements as supporting evidence rather than final competency scores.

There are a number of remaining issues with this present study. Firstly, the observations reported here stem from a five question development interview, which may not be representative of a larger population of candidates. Secondly, the criteria used to classify Beginner, Intermediate and Advanced evidence levels are arbitrary and require further calibration to independent expert annotations. Thirdly, errors in speech transcription may influence the semantic processing of the interview transcript.

Accordingly, the current AEGIE prototype should not be considered a validated tool to assess candidate competency, employment appropriateness, and pass/fail results. A research study with more participants, external validation, calibration of the existing rules, and a different set of responses has to be carried out in order to make conclusive inferences about the efficacy of this framework.

## VIII. CONCLUSION AND FUTURE WORK

In this work, we presented an adaptive AEGIE framework that combines speech transcribing, speech delivery analysis, semantic comparison, adaptive questioning, and feedback to the general workflow of the technical interview. NVIDIA Parakeet transcribes speech into text and we employ MiniLM to analyze semantic correlation across entire answer, sentence, and concept levels. We keep these metrics separate so each aspects of a candidates' answer can be inspected independently without weighting with a single score.

The five-question development experiment tested the entire process, from voice input to evidence presentation. Mean WPM across the recorded responses was 117.33, mean whole-answer similarity was 0.6522, and mean best concept-level similarity was 0.4994. For this set of data, the prototype evidence calculation was 0.5605, which is within the Intermediate range of the current development criteria. These measurements are observations from the prototype experiment, and should not be accepted as affirmed measures of candidate competency.

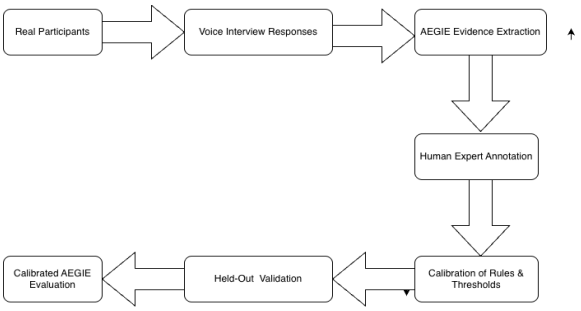



Fig. 3. Human annotation, calibration, and validation AEGIE workflow.

Future work will include gathering responses from human subjects and securing independent annotations from human experts. The evaluation will be based on correctness, relevance, coverage of concepts, and ease of communication. Expert annotations can then be used to adjust the current decision rules and to evaluate the framework against a different set of unseen responses. Other experiments will explore the effects of transcription errors, a larger question bank, speed, and the role of the individual levels of semantic evidence.

What is the purpose of AEGIE? It is designed to help human interviewers, not take their jobs. Since traceability will be built into speech and semantics, the framework can help. It enables the human interviewer to take a systematic look at technical responses.

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